Software Requirements Specification

MindPay — Team Project SRS

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

> *This document uses the instructor's Week 3 SRS structure. The content below describes MindPay and the eight functional requirements selected for the four-month MVP; the original headings, tables, and visual format are retained.*

Field	This project
Team name	13. Socrates and Skeletons
Project title	MindPay — an online personal-finance and spending-insight assistant
SDLC chosen in M1 + reason	Incremental — deliver transaction management and the overview dashboard first; then add voice, slip detection, AI coaching, Money Runway, accountability, and online-state handling in controlled increments across four months
Date / version	v1.2 · 29 Aug 2026 · Four-month MindPay MVP SRS
GitHub repo	Not yet created

1. Introduction

1.1 Purpose and scope

Users often delay expense logging, forget transaction details, and see only a remaining balance without understanding how quickly it is being spent. MindPay is an online personal-finance assistant that combines transaction management, an overview dashboard, voice entry, automatic user-initiated gallery slip detection, an AI Persona Coach, Money Runway, voluntary Anti-Flex Accountability Mode, and a clear online operating state. The four-month MVP contains eight functional requirements. Success means that a user records spending more consistently, understands current financial status, and can see when money may become low without hidden submissions, public shaming, or unverified financial advice.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
User	primary user	Accurate transactions; clear dashboard; quick voice or slip entry; Money Runway and AI guidance	Incorrect extraction, lost input, misleading summaries, or private financial details being exposed
Accountability partner	secondary user	Clear consent, limited budget-status sharing, and the ability to leave at any time	Forced disclosure, insulting alerts, or exact financial amounts being shared without permission
Course instructor / external service provider	stakeholder or supplier	Testable four-month scope, stable interfaces, minimized data, and honest evidence	Data leakage, unexplained AI output, or claims for features that were not built

2. Overall description

2.1 Product context — what it does / does not

MindPay is an online mobile application. It stores user-confirmed transactions, budget settings, and consented accountability status through an authenticated server. External services are used only where needed for speech transcription, slip detection, AI wording, and notifications. Voice results remain editable drafts until the user confirms them. Slip results cannot be edited. Everything below is the complete scope of the four-month MVP; no additional feature is implied.

IN SCOPE (4-MONTH MVP)	OUT OF SCOPE (explicit promise)
• FR-1 Transaction management • FR-2 Overview dashboard • FR-3 Voice transaction logging • FR-4 Automatic gallery slip detection • FR-5 AI Persona Coach • FR-6 Money Runway / estimated low-balance date • FR-7 Anti-Flex Accountability Mode • FR-8 Online operating mode	• Continuous location tracking or gallery monitoring without a user-initiated scan • Group chat or messaging inside the app • Full offline submission or automatic background synchronization • Home-screen widget • Shared wallet • Debt categories

2.2 Assumptions and constraints

- Assumption: every user has a supported smartphone, a valid account, and internet access for submitted operations.
- Constraint: Voice results are drafts, while slip results cannot be edited; the user must confirm or correct extracted values before saving.
- Constraint: the selected operating mode is online; loss of connection may keep an unfinished local draft but never counts it as submitted.
- Constraint: the app stores only data required for transactions, budget summaries, accountability consent, and verification; a gallery scan starts only after an explicit user request, and the app does not continuously monitor photos or location.
- Constraint: financial estimates are decision-support information, not guarantees or investment advice; the eight-FR MVP must be implemented and tested within four months.

3. Functional requirements

FR-1 — Transaction management

User story: As a user, I want to create, edit, categorize, and delete my transactions, so that my financial records remain accurate and organized.

Pain this traces to: incomplete or incorrect transaction records make every summary and forecast unreliable.

1. Given an authenticated user and valid transaction details, when the user confirms a new income or expense, then the system stores one transaction with amount, type, category, date, and note and displays it in history.
2. Given a transaction owned by the user, when the user edits or deletes it and confirms the action, then the history and all dependent summaries reflect the change once.

FR-2 — Overview dashboard

User story: As a user, I want an overview dashboard, so that I can understand my current balance, income, expenses, and category totals at a glance.

Pain this traces to: reviewing separate transaction records makes it difficult to recognize current financial status and spending patterns quickly.

1. Given confirmed transactions in the selected month, when the dashboard loads, then it displays total income, total expenses, current balance, and categorized spending calculated from those records.
2. Given the user changes the dashboard period or a confirmed transaction changes, when the dashboard refreshes, then every displayed total and chart updates to the same selected dataset.

FR-3 — Voice transaction logging

User story: As a user, I want to describe a transaction by voice, so that I can record it without typing every field.

Pain this traces to: repeated manual entry feels slow, so users postpone logging and later forget the amount or category.

1. Given microphone permission and an internet connection, when the user says 'Lunch 60 baht today', then the system shows an editable draft with amount 60 THB, today's date, and a suggested category.
2. Given a voice draft, when the user corrects any field and confirms it, then only the confirmed values are saved and one transaction appears in history.

FR-4 — Automatic gallery slip detection

User story: As a user, I want the app to automatically detect a supported payment slip when I start a gallery scan, so that I do not need to select the slip or retype its details manually.

Pain this traces to: manually searching for and selecting a payment slip, then retyping its details, is repetitive and discourages consistent logging.

1. Given photo-library permission and at least one supported payment slip among the 20 most recent images, when the user starts Scan Gallery, then the system automatically identifies the most recent supported slip without manual image selection and places the available amount, date, time, bank

FR-5 — AI Persona Coach

User story: As a user, I want financial feedback in a communication style I choose, so that spending information is easier to understand and act on.

Pain this traces to: charts show numbers but may not explain why a spending pattern matters or what small action the user can take next.

1. Given the user selects Supportive, Direct, or Reflective tone, when the coach generates feedback, then the message follows that tone, uses only verified values, and is labeled as AI-generated guidance.
2. Given the AI service is unavailable, when the user opens the coach, then the app shows a neutral rule-based summary and still allows transaction recording.

FR-6 — Money Runway / estimated low-balance date

User story: As a user, I want to see how long my available money may last, so that I can adjust spending before the balance becomes too low.

Pain this traces to: a remaining balance does not show how quickly money is being spent, so users may run out before the end of the month.

1. Given an available balance and confirmed expenses from the previous seven days, when the dashboard loads, then the system displays an estimated low-balance date calculated from the seven-day average daily expense.
2. Given the seven-day average daily expense is zero, when the dashboard loads, then the system displays 'No depletion date from current data' and does not divide by zero.

FR-7 — Anti-Flex Accountability Mode

User story: As a user, I want to share a limited budget-status signal with a trusted partner, so that voluntary accountability helps me follow an agreed saving goal.

Pain this traces to: a self-set budget can be easy to ignore when there is no supportive and consented accountability.

1. Send notifications to parents via LINE with the transaction details. Users will see only the approved status—Within Budget, Near Limit, or Over Budget—while exact amounts remain hidden by default.
2. Given either user disables Accountability Mode, when the action is confirmed, then future status alerts stop immediately and the relationship becomes inactive.

FR-8 — Online operating mode

User story: As a user, I want the app to show whether online services are available, so that I know when my changes can be submitted safely.

Pain this traces to: silent connection failures create uncertainty, duplicate attempts, or lost input.

1. Given an internet connection and a valid session, when the user submits a transaction or accountability change, then the server stores it once and the app displays a visible success state.
2. Given no internet connection, when the user selects Submit, then the app displays Offline, does not claim success, and retains the unfinished input as a draft for review and manual submission after reconnection.

4. Non-functional requirements

Forbidden words unless quantified: fast, easy, user-friendly, smart, secure. NFR-5 (security/privacy) is required for M2 — every requirement below has a measurable check.

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	The authenticated dashboard shall load within 2 seconds for an account containing up to 5,000 transactions on the team's reference device and network.	Measure 30 dashboard loads; at least 27 must meet the target.
NFR-2	performance	A voice draft shall appear within 3 seconds after speech ends, and the first clear supported slip draft shall appear within 5 seconds after a user starts Scan Gallery with one supported slip among the 20 most recent images, under normal network conditions.	Run 30 scripted voice trials and 30 labeled gallery-scan trials with one clear supported slip among the 20 most recent images; at least 27 of each must meet its target.
NFR-3	usability	A first-time user shall create and confirm one transaction by voice or slip in 45 seconds or less without assistance.	Run a timed usability trial with at least five users who have not seen the app.
NFR-4	reliability	Retrying a timed-out write request with the same idempotency identifier shall create no more than one server record.	Simulate a timeout and retry the same request three times; verify one stored record.
NFR-5	security / privacy (required)	A user shall access only owned financial records and accountability status explicitly shared through active consent; changing a user or relationship identifier shall not bypass authorization.	Tamper with API identifiers and confirm rejection with HTTP 403 or equivalent.

5. Use cases

5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1 Manage transaction	User	FR-1	User creates, edits, categorizes, or deletes an owned transaction and sees one accurate history.
UC-2 View overview dashboard	User	FR-2	Dashboard shows balance, income, expenses, and categorized totals from the selected period.
UC-3 Record by voice	User	FR-3	System scan an expense, corrects the draft, and confirms one transaction.
UC-4 Scan gallery for slips	User → Slip/OCR service	FR-4	User starts Scan Gallery; the system detects the most recent supported slip automatically.
UC-5 Request coach feedback	User → AI service	FR-5	User receives respectful, labeled feedback based on verified values.
UC-6 View Money Runway	User	FR-6	Dashboard shows an estimated low-balance date and handles zero spending safely.
UC-7 Enable Accountability Mode	Two users	FR-7	Send notifications to parents' LINE chats to inform them of transaction details.
UC-8 Handle lost connectivity	User	FR-8	App marks input as an unsent draft and waits for manual submission online.

5.2 Use-case diagram

> *Not drawn yet on purpose. Week 3's job is naming these eight use cases correctly against the FR list above; the actor/association/<<include>> diagram itself is Week 4's task, taught in the next session.*

6. Out of scope

1. Continuous location tracking or gallery monitoring without a user-initiated Scan Gallery action.
2. Group chat or messaging inside the application.
3. Full offline submission or automatic background synchronization; the selected mode is online.
4. A home-screen widget.
5. A shared wallet.
6. Debt categories and debt-repayment tracking.

7. Traceability matrix (Golden Thread)

Only FR-6 is filled all the way across; this is the single MindPay thread followed from problem to test (Ch3 → Ch4/5 → Ch6 → Ch7). The other rows stay open until their week arrives, exactly as required for M2.

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Records become incomplete or inaccurate	FR-1 transaction management	(Week 5)	(Week 6–7)	(Week 7)
Financial status is difficult to review	FR-2 overview dashboard	(Week 5)	(Week 6–7)	(Week 7)
Slow manual entry	FR-3 voice logging	(Week 5)	(Week 6–7)	(Week 7)
Manual slip search and retyping	FR-4 automatic slip detection	(Week 5)	(Week 6–7)	(Week 7)
Charts lack explanation	FR-5 AI coach	(Week 5)	(Week 6–7)	(Week 7)
Balance hides spending pace	FR-6 Money Runway	Transaction entity + seven-day average calculator (Ch4–5)	Money Runway dashboard card (Ch6)	calculate_money_runway() unit tests (Ch7)
No voluntary accountability	FR-7 accountability	(Week 5)	(Week 6–7)	(Week 7)
Connection failures are unclear	FR-8 online mode	(Week 5)	(Week 6–7)	(Week 7)

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
28 Aug 2026	Gemini	Propose distinctive finance-app features.	Candidate automation and behavioral features.	Selected the eight MVP FRs and moved the widget, shared wallet, debt categories, geofencing, mood, regret, and gacha outside the four-month scope; confirmed Online mode.	Scope must match what the team can build and test.
28 Aug 2026	ChatGPT	Align the instructor's MindPay SRS with the eight FRs in Section 2.1 without redesigning it.	Eight FRs, five NFRs, matching use cases, out-of-scope items, Golden Thread, and self-check wording.	Preserved the original headings and tables; renumbered every dependent section and replaced user terminology with User.	Every section must use the same authoritative FR list.
28 Aug 2026	Claude	Validate whether the eight-FR MVP can be implemented and tested within four months.	A feasibility second opinion covering delivery phases, dependencies, and major risks.	Confirmed an incremental four-month plan: core transactions and dashboard first; voice/slip next; AI, Money Runway, accountability, and online-state handling after the core data flow. Widget, shared wallet, and debt remain out of scope.	The scope is feasible if the team reuses proven speech, OCR, and LLM services and keeps integrations limited.
29 Aug 2026	ChatGPT	Revise FR-4 and UC-4 so the system detects a gallery slip automatically after a user-initiated scan, without expanding the four-month scope.	Revised FR-4 and UC-4 wording plus aligned context, NFR-2, out-of-scope, and traceability text.	Limited each scan to the 20 most recent images; retained editable drafts, the 80% field-confidence rule, user confirmation before saving, and no continuous background monitoring.	Automatic detection can remain privacy-aware and testable when the user initiates a bounded scan.

9. Self-check

- ☑ **All 8 FRs are user stories ending in a benefit tied to a real pain.**
- ☑ **Each FR has 2 observable Given-When-Then acceptance criteria.**
- ☑ **No unquantified vague words appear in §3.**
- ☑ **Five measurable NFRs include the required security / privacy NFR-5.**
- ☑ **Every FR traces to a pain and a row in §7.**
- ☑ **The four-month scope and explicit out-of-scope list match §2.1.**
- ☑ **Eight use cases map one-to-one to FR-1 through FR-8.**
- ☑ **§8 honestly records Gemini, ChatGPT, and Claude requests, drafts, changes, and checks.**

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.